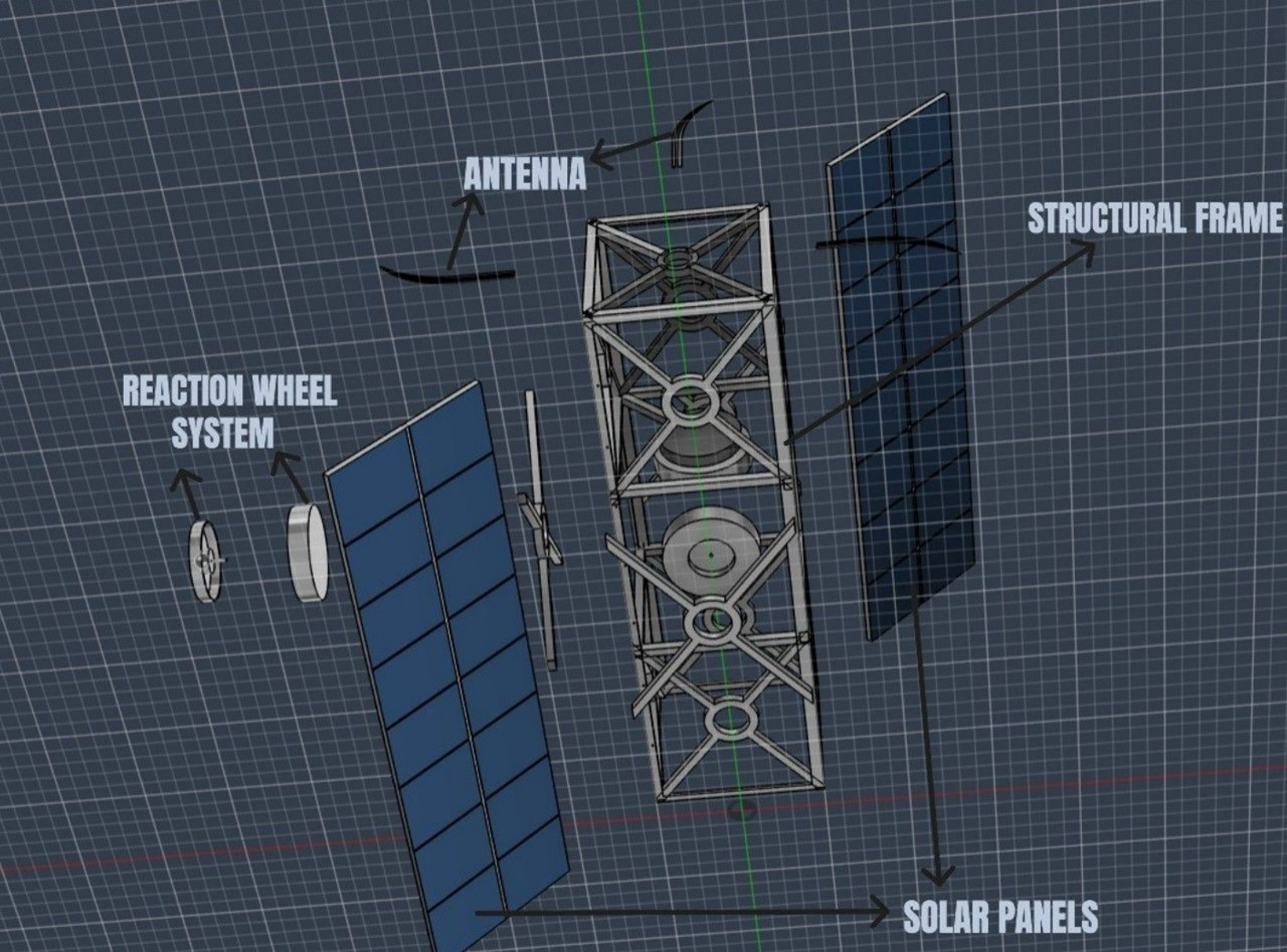


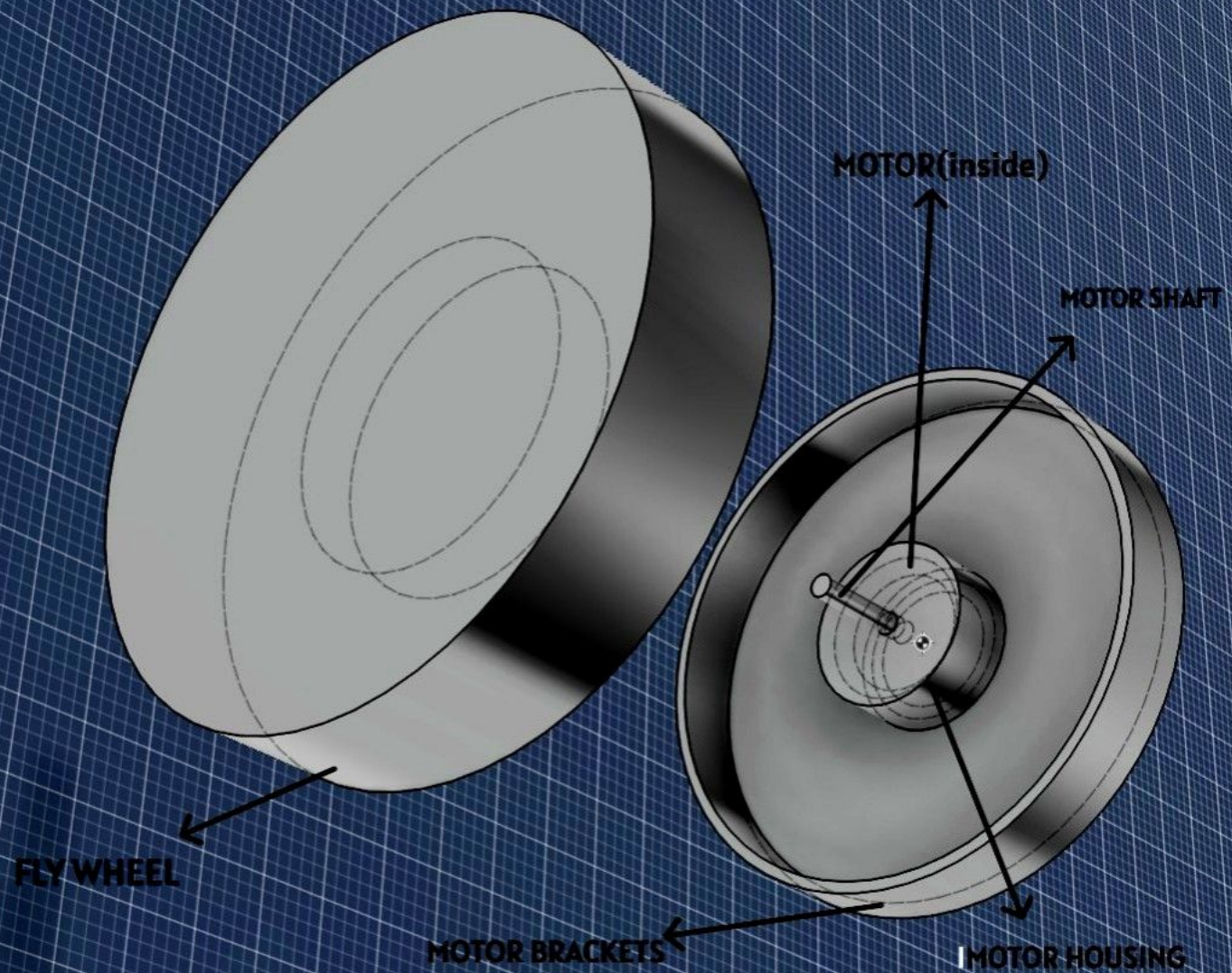
**REACTION WHEEL
SYSTEM**

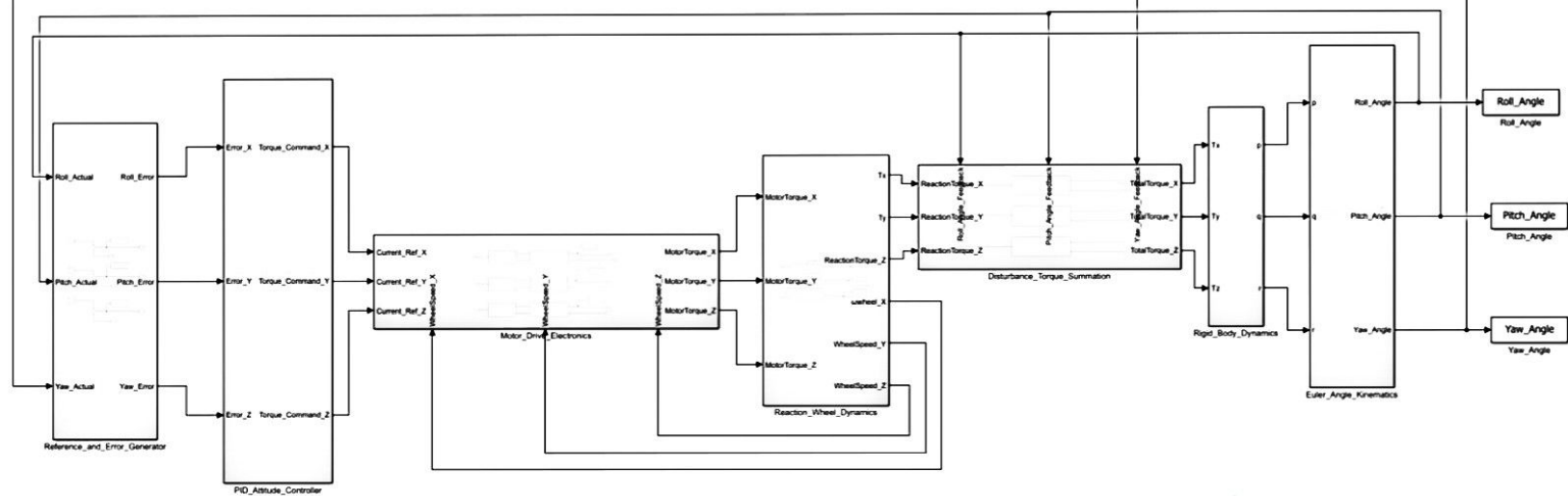
ANTENNA

STRUCTURAL FRAME

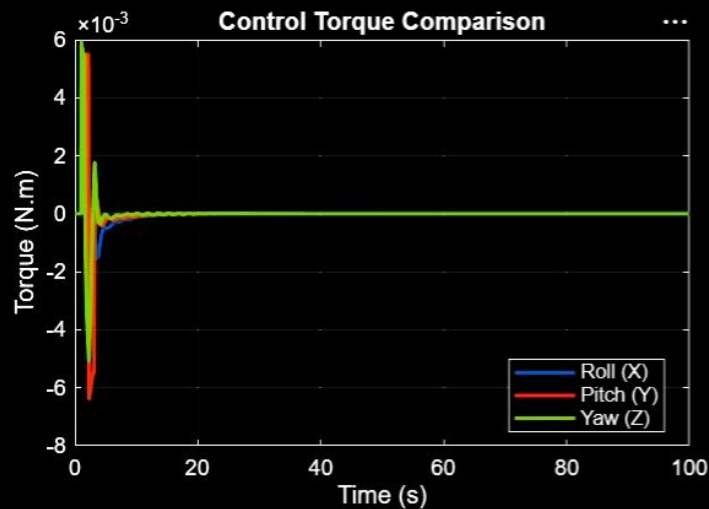
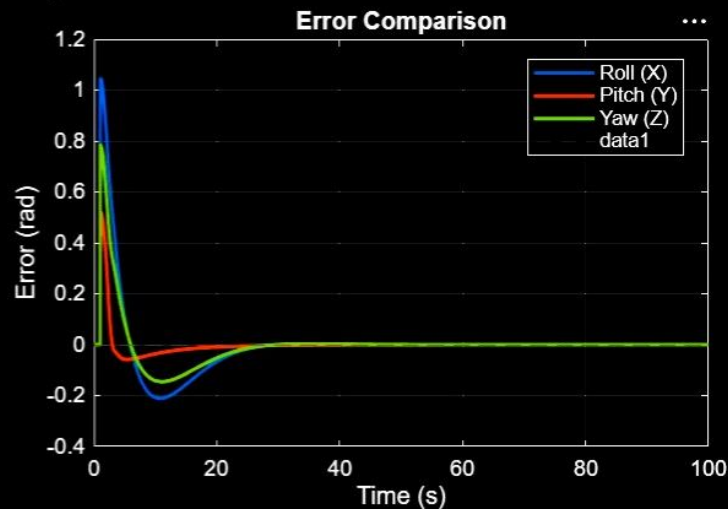
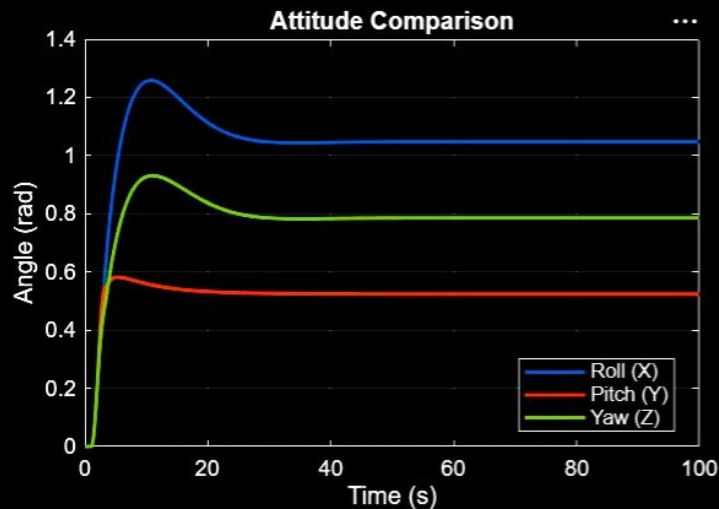
SOLAR PANELS



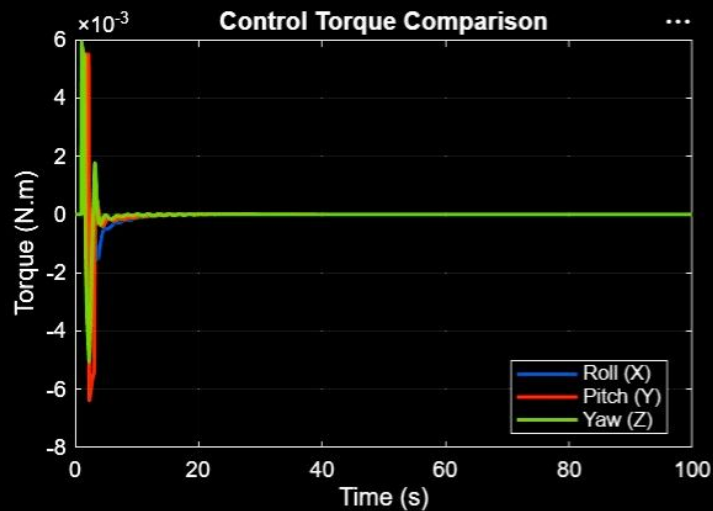
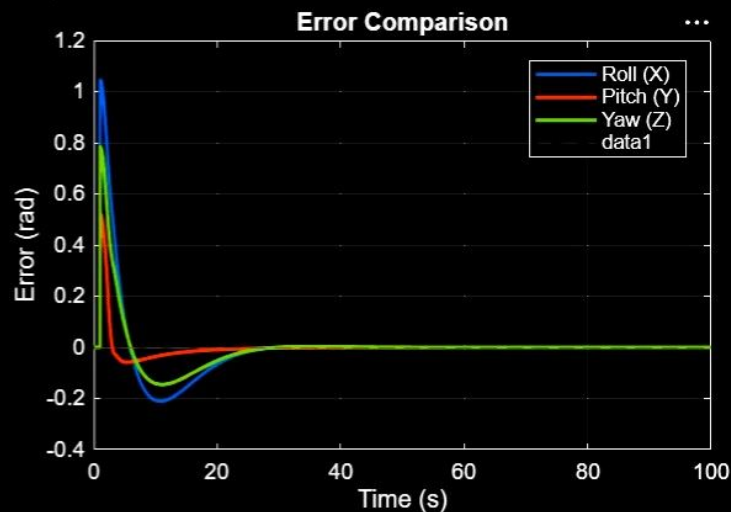
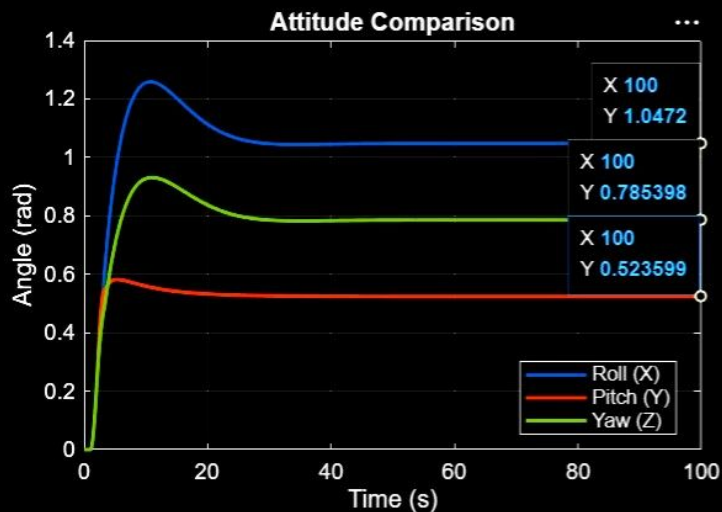




3-Axis Attitude Control -- Comparison Overview



3-Axis Attitude Control -- Comparison Overview



===== 3-AXIS SIMULATION RESULTS =====

Metric	Roll (X)	Pitch (Y)	Yaw (Z)
Desired Angle (deg)	60.000	30.000	45.000
Rise Time (s)	3.327	1.207	3.466
Peak Time (s)	10.793	5.223	11.113
Overshoot (%)	20.17	11.16	18.54
Settling Time (s)	24.353	18.843	24.703
Steady State Err (rad)	0.000000	-0.000000	0.000000
RMS Error (rad)	0.159400	0.061908	0.114131
Max Torque (N.m)	0.005933	0.005933	0.005933
Max Wheel Speed(rad/s)	70.000000	70.000000	70.000000

===== 3-AXIS SIMULATION RESULTS =====

Metric	Roll (X)	Pitch (Y)	Yaw (Z)
Desired Angle (deg)	60.000	30.000	45.000
Rise Time (s)	3.330	1.210	3.460
Peak Time (s)	10.790	5.230	11.120
Overshoot (%)	20.17	11.16	18.54
Settling Time (s)	24.350	18.840	24.700
Steady State Err (rad)	-0.000000	-0.000000	-0.000000
RMS Error (rad)	0.176564	0.071426	0.127097
Max Torque (N.m)	0.005936	0.005936	0.005936
Max Wheel Speed(rad/s)	70.000000	70.000000	70.000000